From Surface Signatures to Selective Lexical Control

Anonymous ACL submission

Abstract

Can a recurrent difference between machine and human prose become a controllable model coordinate? We trace one corpus-conditioned realization signature through four linked tests: surface recurrence, cross-generator hidden readout, fixed-token causal transport, and free-generation realization. The signature is estimated from 500 matched English continuations under two elicitation interfaces and evaluated in Llama 3.1 8B and Qwen3 4B. Lexical and syntactic directions recur across generators and remain continuously readable when the evaluated generator is excluded from fitting. A source-consistent Jacobian pullback then gives lexical transport beyond every one of 100 complete fitted-pipeline nulls in both models ($p = 0.0099$); syntax remains a readable comparison direction. Finally, a 20-prompt development set selects a single intervention magnitude before a held-out test on 166 untouched prompts and three paired decode seeds. In Llama, the lexical intervention produces a target projection of $0.269\,[0.127, 0.412]$ and component selectivity of $0.287\,[0.117, 0.459]$; all three seed means are positive, while syntactic and discourse off-target intervals cover zero. The result connects a distributed surface regularity to selective lexical control under a frozen, explicitly scoped intervention.

1 Introduction

A human literary continuation asks, "Why prematurely claim to have reached finality, when unexpected novelties may shatter any system before it is even completed?" Its matched machine continuation states, "Claiming finality is premature because unexpected novelties can shatter any system before completion." The proposition remains, but a rhetorical question becomes a declarative causal relation and the wording is regularized. We ask whether such recurring choices are merely visible at the surface or can be located and selectively moved inside a language model.

Prior work supplies the endpoints of this question but not the bridge between them. Linguistic analyses identify lexical, syntactic, and rhetorical contrasts in generated text (Zamaraeva et al., 2025; Zanotto and Aroyehun, 2025; Reinhart et al., 2025); representation methods recover linguistic information from model states (Hewitt and Manning, 2019); and activation interventions alter model behavior (Rimsky et al., 2024). A surface regularity, an accessible hidden coordinate, and a realized output effect are nevertheless different objects. Connecting them requires keeping the linguistic construct fixed while the evidence changes from observation to intervention.

We define a *machine–human realization signature*: a corpus-conditioned direction over recurrent wording and clause-structure choices in continuations of the same natural context. The signature is estimated from 500 matched pairs under a proposition-conditioned interface, then challenged on a prefix-only interface that removes the human-derived plan. This design asks whether a distributed realization pattern survives both generator variation and a change in elicitation, rather than turning familiar words into an authorship detector.

Figure 1 turns this question into a four-stage evidence chain. We first establish recurrence of frozen lexical and syntactic axes across generators, domains, and interfaces. We then train continuous readouts while withholding an entire generator and require prediction on both human and machine continuations. Next, a source-consistent Jacobian pullback maps the late lexical readout to an early intervention and compares its five-checkpoint transport curve with 100 null pipelines that repeat every fitting step after permuting only the linguistic correspondence. Finally, we freeze the lexical pole and intervention site, select one magnitude on 20 development prompts, and test target-aligned,

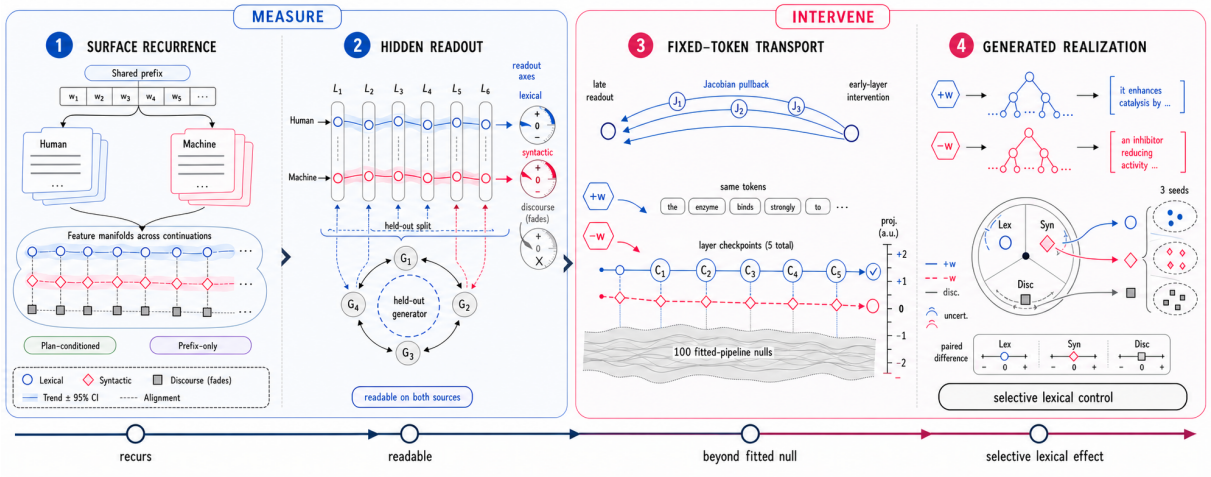


Figure 1: **From recurrence to selective realization.** Frozen surface axes are read across held-out generators, pulled back through model depth, tested against 100 refitted nulls, and evaluated in paired generation.

component-selective realization on 166 untouched prompts with three paired seeds.

The lexical direction passes all four stages. It recurs under both interfaces, remains readable for every held-out generator, and exceeds all refitted transport nulls in both Llama and Qwen. In the final Llama test, the $+10\%/-10\%$ intervention shifts the lexical score by $0.269\,[0.127, 0.412]$ and exceeds the mean off-target response by $0.287\,[0.117, 0.459]$. The effect is positive in every seed; its target-minus-syntax and target-minus-discourse contrasts are also positive. Thus the same lexical construct is observable at the surface, readable in hidden states, causally transportable on fixed sequences, and selectively realizable in sampled text.

The contribution is deliberately specific. The final selective-generation result concerns the lexical axis in Llama at a development-selected single-layer strength; syntax remains a comparison axis and the earlier 5% multi-condition audit is reported as sensitivity evidence. Within that scope, the result provides an end-to-end account of how a distributed linguistic signature can become a controllable coordinate rather than stopping at detection or probing.

Table 1 summarizes the changing evaluation unit from held-out generators to fixed sequences, refitted pipelines, and held-out prompt-level effects.

2 Related Work

2.1 Surface Signals

Machine and human text differ in vocabulary, dependency structure, and discourse organization, although the reported cues vary across models and domains (Muñoz-Ortiz et al., 2024; El Attar et al., 2026; Ahmad et al., 2026). Detector failures under distribution shift show why these cues should not be treated as universal authorship rules (Sadasivan et al., 2024; Li et al., 2024; Shamardina et al., 2025). We instead define a corpus-conditioned multivariate direction and ask whether it recurs across named generators, domains, and elicitation interfaces.

2.2 Representation Readout

Representation-based detectors can separate human and generated text from observer-model states (Chen et al., 2025; Vishnyakov and Gaintseva, 2026). Our readout carries a different burden: it predicts an independently measured continuous linguistic score on both sides of the source contrast, with the evaluated generator excluded from every fit. Positive human/machine agreement is required before the two source-side patterns form a consensus direction, so a source-only boundary cannot by itself satisfy the test.

2.3 Causal Intervention

Probes test whether information is accessible, whereas causal use requires intervention evidence (Hewitt and Manning, 2019). Activation addition controls varied behaviors (Turner et al., 2023; Rimsky et al., 2024; Li et al., 2023), but reliability depends on readout–behavior alignment, specificity, and direction identifiability (Canby et al., 2025; Waldis et al., 2026; Goyal and Daumé III, 2026; Venkatesh and Kurapath, 2026). Nullspace and amnesic methods further show that an intervention's meaning depends on the variation it removes (Ravfogel et al., 2020; Elazar et al., 2021). Together,

Stage	Estimand	Construction and evaluation
Surface	Axis u_g	500 paired feature differences; Δ, paired d, and FDR q.
Readout	Cross-generator D	Three-generator train, one-generator test; Pearson r on both sources.
Pullback	Canonical w	32 balanced Jacobian pairs; source-side agreement and frozen-anchor response.
Trace	Capacity C	Five intermediate checkpoints on 100 identical fixed sequences.
Null test	Curve p_{null}	100 complete pipeline refits and an add-one curve-statistic rank.
Generation	Target p_g	Paired $+\alpha/-\alpha$ generations; strength selected on 20 development prompts and tested on 166 untouched prompts.
	Selectivity b_g	Target response minus the mean of two off-target responses.

Table 1: **Evidence-chain estimands.** Units progress from held-out generators to fixed sequences, refitted pipelines, and held-out prompt effects.

these literatures motivate separate tests of surface recurrence, hidden readability, fixed-token transport, and generated-language realization. We apply those tests in sequence to one fixed linguistic construct.

Our setting therefore sits between computational stylistics and mechanistic intervention. Surface studies supply a measurable linguistic construct, but distribution-shift failures caution against turning it into a detector; representation methods supply candidate coordinates, but accessibility alone does not establish functional use; steering methods supply an intervention, but output change must be aligned with the construct that motivated it. Keeping the same score fixed across these interfaces lets each literature contribute a distinct evidential link instead of allowing success at one link to rename the claim at the next.

3 Method

3.1 Surface Signature

Two interfaces. The proposition-conditioned corpus contains 500 English continuation pairs balanced across PG-19, WikiLingua, Wikipedia, and XL-Sum (Rae et al., 2020; Ladhak et al., 2020; Wikimedia Foundation, 2023; Hasan et al., 2021). A planner extracts a proposition plan from each human continuation, a distinct generator continues the natural prefix from that plan, and a third model checks coverage and contradiction. The prefix-only corpus regenerates the same 500 samples from the prefix and target length alone. Mean semantic cosine is 0.925 under proposition conditioning and 0.727 under prefix-only generation, compared with 0.469 after within-source derangement. The proposition-conditioned interface provides stronger content control, whereas the prefix-only interface removes the asymmetric plan-to-text channel. Section C.2 records construction, rejection, and leakage audits.

The two arms are not interchangeable replications; they expose opposite sources of uncertainty. Proposition conditioning makes content mismatch a less plausible explanation, but the machine alone receives the explicit plan and may lexicalize its structure. Prefix-only generation removes that route, but allows larger semantic divergence. We therefore fit every axis and readout only on the proposition-conditioned arm and treat prefix-only text as an untouched transfer environment. Recurrence across the two arms cannot establish content-invariant authorship, but it is harder to attribute solely to either plan verbalization or unconstrained topic drift.

Sample selection is frozen before linguistic measurement. All 500 accepted proposition-conditioned pairs pass an independent automatic judge, use distinct planner, generator, and judge models, and receive no post-generation repair. Their machine–human length ratio has mean 0.935 and median 0.925, with range 0.603–1.405; length and six further nuisance statistics are nevertheless removed inside every training fold. These controls do not certify rhetorical equivalence, but they prevent the surface construct from being defined by obvious selection, repair, or length asymmetries.

Frozen directions. We predefine 35 measurements: 10 lexical, 12 syntactic, and 13 discourse features. Seven length, entity, and number statistics are nuisance variables rather than target features. Each training fold residualizes the linguistic measurements against those nuisances, standardizes them, and projects them onto the paired machine-minus-human direction. Lexical and syntactic axes are confirmatory targets; discourse remains an off-target audit because it fails the same stability gate. The full dictionary, fold construction, PCA sensitivity, and same-token vocabulary analysis are reported in Sections C.3 and C.4.

The construction treats the signature as a distributed relation rather than a list of diagnostic

words. Grouping measurements before analysis prevents a conspicuous cue from defining the target after outcomes are known, while fold-local nuisance removal prevents length, entity density, or number density from becoming shortcuts. The paired direction then asks which combination of residualized choices changes consistently between continuations of the same context. This design makes lexical and syntactic scores interpretable as degrees of realization within the observed corpus; it does not convert any individual feature into an authorship rule.

3.2 Hidden Readout

Let h_i^k be the mean continuation state after block k, projected away from the leading ten pooled within-class nuisance directions. For each held-out generator and source side, we fit a forward covariance pattern between these states and the frozen surface score. Human and machine patterns must have positive cosine before their normalized sum defines the cross-generator readout v_g^k. Every nuisance fit, surface direction, and hidden pattern excludes the evaluated generator. The readout is successful only if it predicts the continuous score on both human and machine continuations, rather than merely separating their sources.

$$\begin{aligned} \widetilde{v}_{ga}^k &= H_a^\top z_{ga}/(z_{ga}^\top z_{ga}), \quad & v_{ga}^k &= \mathrm{unit}(\widetilde{v}_{ga}^k), \\ v_g^k &= \mathrm{unit}(v_{g,h}^k + v_{g,m}^k), & a &\in \{h, m\}. \end{aligned} \tag{1}$$

The readout stage tests generalization of the declared linguistic variable, not merely separability of human and machine sources. A source classifier could exploit a generator-specific artifact while remaining unrelated to the frozen realization score. Continuous prediction within both source sides instead asks whether the same hidden direction orders human and machine continuations by independently measured realization, and excluding the evaluated generator asks whether that relation survives a new generator environment. Positive held-out correlations therefore establish accessibility of the declared surface variable under generator transfer; they do not yet show that an early intervention transports it through the model.

The opening literary pair interprets the surface construct, while the held-out continuous test establishes its hidden-state generalization. This division of labor is important: examples reveal what movement along the direction looks like, and generator-level holdout verifies that the readout orders many unseen human and machine continuations by the same independently measured score.

3.3 Fixed-Token Transport

Earlier residual streams use different coordinates from the late readout. For source side a, we translate the readout backward through the layer-to-layer Jacobian and combine the independently estimated human and machine pullbacks only after projecting through the frozen nuisance-nullspace projector P_m:

$$\begin{aligned} a_{ig}^a &= (J_i^{\ell \to k})^\top v_g^k, \\ \hat{a}^a &= \mathrm{unit}(P_m \bar{a}^a), \\ w_{mg} &= \mathrm{unit}(\hat{a}^{\textsc{Human}} + \hat{a}^{\textsc{Machine}}). \end{aligned} \tag{2}$$

Applying the late readout vector directly at an early layer would assume that residual coordinates retain the same meaning across depth. The Jacobian transpose instead maps sensitivity of the declared late readout into the early residual space, so each fitted sequence contributes a local first-order pullback rather than an untransformed activation direction. Human and machine pullbacks are estimated separately, projected through the same frozen nuisance nullspace, and combined without outcome-dependent sign correction. This construction yields a preoriented early direction for the fixed-token test, but it does not optimize the surface form of sampled continuations.

The source-side signs are retained, so $+w$ is oriented by the machineward readout before any generation outcome is observed. On an unchanged token sequence, symmetric $+w/-w$ intervention measures the signed response of the late readout; its absolute value is the transport capacity. The confirmatory statistic is the depth-normalized area over five intermediate checkpoints, excluding the injection layer and fitted readout anchor because both are construction-internal.

$$\begin{aligned} \tau_{img}(w) &= \frac{(h_i^k(+\alpha w) - h_i^k(-\alpha w))^\top v_g^k}{2\alpha \|w\|_2}, \\ \widehat{C}_{mgj} &= \frac{1}{n} \sum_i |\tau_i(w_{mg}, v_g^j)|, \\ \widehat{C}_{mgj}^0 &= \frac{1}{R} \sum_r \left[\frac{1}{n} \sum_i |\tau_i(w_{mgr}, v_{gr}^j)| \right]. \end{aligned} \tag{3}$$

The comparison is a complete fitted-pipeline null, not a random input direction. Each of $R = 100$ replicates permutes surface scores within

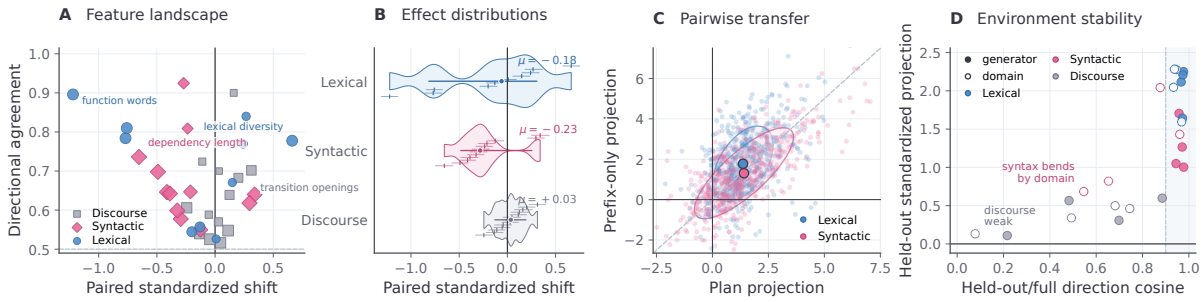


Figure 2: **Lexical and syntactic directions recur across environments.** (A) Feature shifts. (B) Group effects. (C) Interface transfer. (D) Generator and domain holdouts.

corpus–generator strata, refits every layerwise readout, reconstructs the source-consistent pullback, and intervenes on the same 100 hashed sequences. The add-one rank $p = (1 + \#\{T_r^0 \geq T\})/(R+1)$ asks whether the learned depth profile is unusual when every fitting advantage is retained but the linguistic correspondence is broken. Section D gives the complete estimands, layer conventions, trajectory checks, and sensitivity settings.

3.4 Generated-Language Realization

Fixed-token transport measures a late readout along an unchanged trajectory; free generation asks whether that coordinate survives autoregressive feedback. We separate strength selection from evaluation. A disjoint 20-prompt development set (indices 0–19) compares candidate magnitudes while keeping the lexical pole, L7 intervention site, token budget, and decoding rule fixed. The selected endpoint, $\alpha = 2$, equals 10% of the mean hidden-state norm. We then freeze the complete protocol and evaluate prompts 34–199: 166 untouched prompts under three paired decode seeds. For measurement group g, target projection p_{ig} and component selectivity b_{ig} are

$$p_{ig}(w) = s_g(Y_i^{+w}) - s_g(Y_i^{-w}),$$
$$b_{ig}(w) = p_{ig}(w) - \frac{1}{2}\sum_{c \neq g} p_{ic}(w). \quad (4)$$

The formal test uses 128 new tokens, temperature 0.7, top-$p = 0.95$, and base seed 8236. For each prompt, we compute the paired $+\alpha/-\alpha$ difference in each seed, average the three differences within prompt, and bootstrap the 166 prompt means. The earlier 5% one-seed matrix and multi-condition replication are retained in Section E.1 as low-dose sensitivity evidence rather than used to estimate the formal effect.

Target projection asks whether the output moves along the declared lexical axis; component selectivity asks whether that movement exceeds the mean syntactic and discourse response. We additionally test lexical-minus-syntactic and lexical-minus-discourse contrasts separately and report seed means alongside prompt-clustered intervals. This hierarchy makes selective realization a joint claim: the target effect, aggregate selectivity, and both pairwise contrasts must point in the declared direction.

4 Experiments

Models and splits. We evaluate hidden readout and intervention in Llama 3.1 8B and Qwen3 4B. Surface axes and readouts are fitted only on the proposition-conditioned corpus. Generator transfer follows a four-fold leave-one-generator-out design, while the prefix-only corpus remains untouched until transfer evaluation.

Evaluation protocol. Readout is evaluated separately on held-out human and machine continuations. Fixed-token transport uses 100 identical sequences, five intermediate checkpoints, and 100 complete fitted-pipeline null refits. Generated-language realization selects the lexical strength on 20 development prompts, then uses 166 disjoint test prompts with paired $\pm 2w$ outputs, three decode seeds, and prompt-clustered bootstrap intervals. Table 1 summarizes the estimands; Sections D and E.1 give the frozen settings.

5 Results

Surface recurrence. Figure 4 separates feature magnitude from cross-environment stability. The six largest paired effects span both confirmatory groups; lexical and syntactic directions remain aligned under every generator holdout and on the untouched prefix-only corpus. All three group-level sign-flip tests yield $p = 0.0002$; prefix-only transfer is confirmatory for the lexical and syntactic

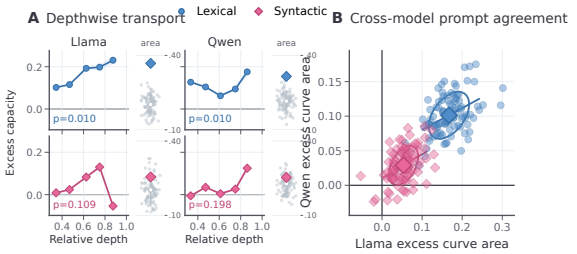


Figure 3: **Lexical transport exceeds the fitted-pipeline null in both models.** (A) Five-checkpoint excess capacity. (B) Prompt-level excess area.

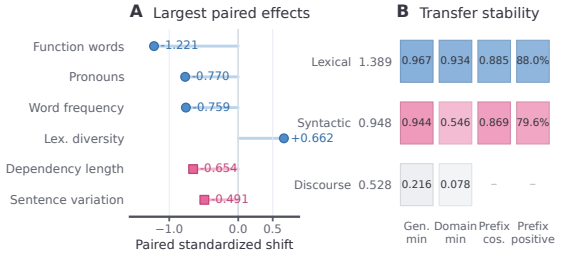


Figure 4: **Surface recurrence is distributed and environment-tested.** (A) Six largest paired shifts. (B) Group shift, worst generator/domain agreement, and prefix-only transfer. Circles are lexical and squares syntactic.

axes. Syntax bends more under domain holdout, while discourse is unstable and remains off-target. Figure 8 grounds the directions in paired excerpts, and Figure 7 reports the plan-removal transfer.

The abstract direction has visible linguistic content without reducing to a shibboleth. Across the matched corpus, *however* occurs in 83 machine continuations and 34 human continuations, and *despite* in 36 and 7. The evaluative forms *crucial* and *significant* each occur in 22 machine documents but 3 human documents; *stated that* and *noted that* occur in 16 and 13 machine documents and no matched human document. These counts help interpret the multivariate projection, but the direction is defined by the frozen distributed inventory rather than any one expression.

The two transfer tests address different failure modes. Generator holdout within the proposition-conditioned arm asks whether the direction depends on one generator while content is closely matched; transfer to prefix-only outputs asks whether it persists after the human-derived plan is removed. Because prefix-only text enters neither axis construction nor readout fitting, its positive projection and readout correlation cannot result from adapting either estimator to that interface. Agreement across the arms therefore supports recurrence under two opposing content-control regimes within the environments measured here. The syntactic domain minimum of 0.546 and weaker discourse stability keep this conclusion group- and corpus-conditioned rather than universal.

Stable axes. Lexical and syntactic domain-minimum cosines remain positive at 0.934 and 0.546, and 88.0% and 79.6% of prefix-only pairs project machineward on the original axes. The wider syntactic range shows that clause packaging is more sensitive to domain than the lexical combination, even though both transfer across generators and interfaces. Discourse behaves differently: it collapses in its weakest environments and under aggressive PCA truncation. Treating discourse as an off-target audit is therefore part of the result, not a post hoc omission of an inconvenient third axis.

Same-token lexical field. A complementary analysis projects 2.44–2.48 million continuation-token states per model and compares an exact token only when it appears on both sides of a matched pair, after removing each continuation's mean coordinate. This design removes unequal token frequency and the global source offset from the local contrast. Of 612–704 eligible token contrasts per model, 89–379 survive BH-FDR together with generator and domain sign-agreement gates. The field is model- and tokenizer-specific, but it shows that the lexical direction is not exhausted by the machine side choosing a different vocabulary (Section C.4).

Cross-generator readout. All 16 held-out-generator readout observations are positive (Figure 5A and Table 2). Mean proposition-conditioned correlations range from 0.395 to 0.450. Although no prefix-only samples enter fitting, the same readouts obtain mean correlations of 0.421–0.544 on prefix-only machine text. Human/machine pullback cosines are 0.916–0.942, whereas pullback-/encoding cosines are 0.193–0.265. This difference is consistent with related but distinct encoding and pullback coordinates; it is not itself evidence of transport.

Readout strength, not sign, varies: source-side r ranges from 0.313 (Qwen-machine syntax under Qwen holdout) to 0.632 (Qwen-human syntax under MiniMax holdout). Fold/full cosines remain within 0.786–0.941, while human/machine agreement spans 0.325–0.607. The coordinate

Held-out gen.	Axis	$\|\Delta u\|_2$	Llama 3.1 8B (L7→L32)					Qwen3 4B (L8→L36)				
			Human r	Machine r	Mean r	Agree	Stab.	Human r	Machine r	Mean r	Agree	Stab.
MiniMax	Lex.	1.473	0.346	0.464	0.405	0.519	0.855	0.355	0.388	0.372	0.325	0.939
DeepSeek	Lex.	1.435	0.530	0.401	0.466	0.535	0.895	0.459	0.423	0.441	0.487	0.941
GLM	Lex.	1.513	0.545	0.526	0.536	0.524	0.846	0.459	0.592	0.525	0.468	0.936
Qwen	Lex.	1.213	0.399	0.392	0.395	0.486	0.882	0.432	0.426	0.429	0.468	0.925
MiniMax	Syn.	1.590	0.381	0.360	0.371	0.607	0.852	0.632	0.340	0.486	0.553	0.883
DeepSeek	Syn.	1.453	0.337	0.446	0.392	0.433	0.910	0.445	0.520	0.483	0.442	0.910
GLM	Syn.	1.429	0.356	0.396	0.376	0.576	0.786	0.496	0.374	0.435	0.513	0.914
Qwen	Syn.	1.328	0.550	0.335	0.442	0.480	0.911	0.478	0.313	0.396	0.563	0.927

Table 2: **Every held-out generator remains readable on both source sides.** Each fold trains on three generators and reports Pearson r for human and machine continuations from the fourth. Agree and Stab. denote source-side and fold/full direction cosine.

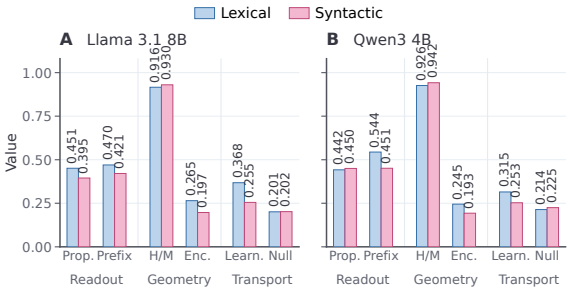


Figure 5: **Readout transfer and null-tested transport are distinct.** Llama (A) and Qwen (B) compare lexical and syntactic results across readout, direction alignment, and transport area.

is therefore orientation-stable under source-side variation, supporting cross-environment readability rather than invariant effect size.

The all-positive pattern is not carried by one easy generator or one source side. Across the complete fold ledger, held-out human- and machine-side correlations range from 0.313 to 0.632, and every generator yields a positive value on both sides for both confirmatory axes. The magnitude changes with generator, source side, and axis, but the sign does not. This is the generalization needed for a consensus readout: the coordinate need not fit every environment equally well, yet it must continue to order independently measured realization after the evaluated generator is removed from every fitting step.

Lexical transport. Llama lexical depth-normalized area is 0.368 against a null mean of 0.201; Qwen is 0.315 against 0.214. No complete null pipeline reaches either learned curve, yielding $p = 1/101 = 0.0099$ in both models. Learned-minus-null intervals are positive at every intermediate lexical checkpoint, and all 1,000 model–layer–sequence signed responses point machineward. This comparison preserves readout fitting, Jacobian translation, nuisance projection, and intervention trajectories; only the linguistic score-to-example correspondence is broken. The lexical effect therefore depends on that correspondence rather than on pipeline fitting alone.

Transport contrast. Syntactic signed response points machineward in 999 of 1,000 cases, and the source-side pullbacks agree strongly, but its depth-normalized areas of 0.255 in Llama and 0.253 in Qwen fall within the corresponding fitted-pipeline null distributions ($p = 0.1089$ and $p = 0.1980$). The contrast identifies lexical realization as the component with correspondence-specific transport in both models. Construction alternatives, nuisance ranks, and fit counts appear in Figure 9.

Because lexical and syntactic axes pass the same surface and readout tests and enter the same intervention pipeline, their divergence isolates the role of the learned score-to-example correspondence. Every refitted null remains below the lexical curve, whereas 10 Llama and 19 Qwen null curves reach or exceed the syntactic statistic. The fitted null therefore selects a cross-model lexical trace rather than rewarding generic directional responsiveness.

Construction robustness. In all four model–axis cases, the canonical pullback has greater five-checkpoint capacity than upstream encoding, raw and nuisance-nullspace source contrasts, and the projected-random ensemble. Nuisance-rank and Jacobian-fit-count variants remain close to the frozen direction. These checks support a source-consistent pullback over simpler substitutes; the complete refitted null still determines the lexical headline result (Figure 9).

Selective generation. The held-out Llama lexical test completes the chain (Figure 6 and Table 3). The target projection is $0.269\,[0.127, 0.412]$, with seed means 0.123, 0.267, and 0.416.

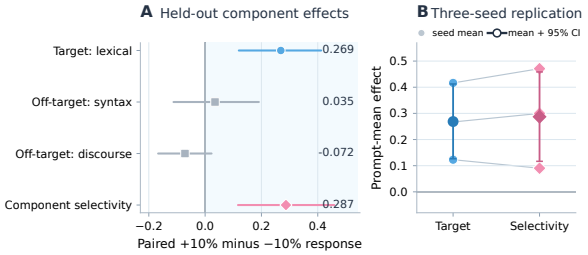


Figure 6: **Held-out lexical control is target-aligned and selective.** (A) Target, off-target, and selectivity intervals. (B) Positive target and selectivity means in all three seeds.

Estimand	Mean [95% CI]	d	p
Lexical target	0.269 [0.127, 0.412]	0.280	0.0006
Syntactic off-target	0.035 [–0.112, 0.191]	0.035	0.6613
Discourse off-target	–0.072 [–0.167, 0.023]	–0.114	0.1454
Component selectivity	0.287 [0.117, 0.459]	0.255	0.0012
Target – syntax	0.234 [0.011, 0.457]	0.162	0.0374
Target – discourse	0.340 [0.184, 0.496]	0.325	< 0.0001

Table 3: **Held-out Llama lexical intervention.** Three-seed prompt means over 166 test prompts; p is a paired sign-flip test.

Component selectivity is $0.287\,[0.117, 0.459]$, with seed means 0.090, 0.300, and 0.471. Syntactic $0.035\,[-0.112, 0.191]$ and discourse $-0.072\,[-0.167, 0.023]$ responses remain centered near zero. The target also exceeds syntax by $0.234\,[0.011, 0.457]$ and discourse by $0.340\,[0.184, 0.496]$. The prompt geometry and smoothed distributions appear in Figure 10.

Visible lexical change. The aggregate projection corresponds to a concrete shift in one of the strongest source features. Relative to $-2w$, $+2w$ lowers function-word rate by $0.0084\,[0.0038, 0.0130]$ in magnitude, matching the machineward sign of the observational effect ($d = -1.221$). Pronoun rate and mean content-word frequency move in the same declared direction on average but have intervals crossing zero. The intervention therefore changes a distributed lexical score with one individually resolved realization, rather than depending on a single trigger word.

6 Discussion

A controllable lexical coordinate. The central result is an end-to-end correspondence: a distributed lexical pattern recurs under two elicitation interfaces, is continuously readable after generator holdout, transports beyond refitted nulls in two models, and selectively changes held-out Llama generations. The same frozen surface score defines every stage, so the final output effect is interpretable as movement of the original linguistic construct rather than an unrelated behavioral change. Development-set strength selection also shows why intervention scale belongs to the controller specification: the 10% endpoint is fixed before the formal prompts and then replicated across three seeds.

What the chain identifies. The four stages progressively narrow the object of study. Surface recurrence identifies a corpus-level realization pattern; held-out readout locates an accessible coordinate that generalizes across generators; the fitted-pipeline null attributes transport to the learned linguistic correspondence; and paired generation establishes that the lexical response exceeds two measured off-target components. No stage is replaced by the next. Their conjunction is what permits a causal control interpretation while keeping the target anchored to observable language.

Scope and implication. Cross-model evidence is strongest through fixed-token transport; selective free-generation evidence is currently specific to the Llama lexical controller. Syntax remains readable but falls inside its fitted-null transport distribution, indicating that the method isolates component-dependent geometry rather than a generic source direction. This separation suggests a practical evaluation recipe for linguistic steering: define the surface construct first, verify held-out readout, test the complete learned pipeline against refitted nulls, choose intervention scale on disjoint development data, and reserve untouched prompts for target and off-target realization tests.

7 Conclusion

We connect a corpus-conditioned lexical signature to selective generation. The direction recurs across generators and interfaces, remains readable after generator holdout, and transports beyond every refitted null in both Llama and Qwen. With strength selected on disjoint development prompts, the frozen Llama controller produces positive held-out target and component-selectivity intervals across three seeds.

References

Zainab Ahmad, Miguel Torres-Ruiz, Ahmad Mahmood, Rolando Quintero, Iqra Ameer, and Necva Bölücü. 2026. Human or machine? a survey on machine-generated text detection. *IEEE Access*, 14:34113–34136.

Marc E. Canby, Adam Davies, Chirag Rastogi, and Julia Hockenmaier. 2025. How reliable are causal probing interventions? In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 857–878. Association for Computational Linguistics.

Xin Chen, Junchao Wu, Shu Yang, Runzhe Zhan, Zeyu Wu, Ziyang Luo, Di Wang, Min Yang, Lidia S. Chao, and Derek F. Wong. 2025. RepreGuard: Detecting LLM-generated text by revealing hidden representation patterns. *Transactions of the Association for Computational Linguistics*, 13:1812–1831.

Yassir El Attar, Esra Dönmez, Maximilian Maurer, and Agnieszka Falenska. 2026. A systematic analysis of linguistic features in AI-generated text detection across domains and models. *arXiv preprint arXiv:2606.04177*.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. 2021. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175.

Navita Goyal and Hal Daumé III. 2026. Steering safely or off a cliff? rethinking specificity and robustness in inference-time interventions. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5723–5738. Association for Computational Linguistics.

Tahmid Hasan, Abhik Bhattacharjee, Md. Saiful Islam, Kazi Mubasshir, Yuan-Fang Li, Yong-Bin Kang, M. Sohel Rahman, and Rifat Shahriyar. 2021. XL-sum: Large-scale multilingual abstractive summarization for 44 languages. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 4693–4703. Association for Computational Linguistics.

John Hewitt and Christopher D. Manning. 2019. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4129–4138. Association for Computational Linguistics.

Faisal Ladhak, Esin Durmus, Claire Cardie, and Kathleen McKeown. 2020. WikiLingua: A new benchmark dataset for cross-lingual abstractive summarization. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4034–4048. Association for Computational Linguistics.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, volume 36, pages 41451–41530.

Yafu Li, Qintong Li, Leyang Cui, Wei Bi, Zhilin Wang, Longyue Wang, Linyi Yang, Shuming Shi, and Yue Zhang. 2024. MAGE: Machine-generated text detection in the wild. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 36–53. Association for Computational Linguistics.

Alberto Muñoz-Ortiz, Carlos Gómez-Rodríguez, and David Vilares. 2024. Contrasting linguistic patterns in human and LLM-generated news text. *Artificial Intelligence Review*, 57(10):265.

Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. 2020. Compressive transformers for long-range sequence modelling. *International Conference on Learning Representations*.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. 2020. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256. Association for Computational Linguistics.

Alex Reinhart, Ben Markey, Michael Laudenbach, Kachatad Pantusen, Ronald Yurko, Gordon Weinberg, and David West Brown. 2025. Do LLMs write like humans? variation in grammatical and rhetorical styles. *Proceedings of the National Academy of Sciences*, 122(8):e2422455122.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics.

Vinu Sankar Sadasivan, Aounon Kumar, Sriram Balasubramanian, Wenxiao Wang, and Soheil Feizi. 2024. Can AI-generated text be reliably detected? In *International Conference on Learning Representations*.

Tatiana Shamardina, Marat Saidov, Alena Fenogenova, Aleksandr Tumanov, Alina Zemlyakova, Anna Lebedeva, Ekaterina Gryaznova, Tatiana Shavrina, Vladislav Mikhailov, and Ekaterina Artemova. 2025. CoAT: Corpus of artificial texts. *Natural Language Processing*, 31:150–175.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Steering language models with activation engineering. *Preprint*, arXiv:2308.10248.

Sohan Venkatesh and Ashish Mahendran Kurapath. 2026. On the non-identifiability of steering vectors in large language models. *Preprint*, arXiv:2602.06801.

Mikhail Vishnyakov and Tatiana Gaintseva. 2026. SV-Detect: AI-generated text detection with steering vectors. *Preprint*, arXiv:2606.07313.

Andreas Waldis, Vagrant Gautam, Anne Lauscher, Dietrich Klakow, and Iryna Gurevych. 2026. Aligned probing: Relating toxic behavior and model internals. *Transactions of the Association for Computational Linguistics*, 14:271–291.

Wikimedia Foundation. 2023. Wikimedia wikipedia dataset, 20231101 english snapshot. Hugging Face Datasets.

Olga Zamaraeva, Dan Flickinger, Francis Bond, and Carlos Gómez-Rodríguez. 2025. Comparing LLM-generated and human-authored news text using formal syntactic theory. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9041–9060. Association for Computational Linguistics.

Sergio E. Zanotto and Segun Aroyehun. 2025. Linguistic and embedding-based profiling of texts generated by humans and large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 22841–22858. Association for Computational Linguistics.

A Limitations

The signature is English and corpus-conditioned. Fixed-token transport is tested in two models, while selective free-generation evidence is limited to the Llama lexical axis at one development-selected, single-layer 10% strength. The result is not a universal authorship factor, does not extend automatically to syntax or Qwen generation, and does not establish semantic or perceptual equivalence.

B Ethical Considerations

The directions are corpus contrasts, not authorship detectors. Using them to label writers may disadvantage non-native, neurodivergent, formal, or technical prose and should not support accusations or penalties. The release contains no detector or decision threshold; any use requires human review and an appeal route.

C Corpus and Surface Evidence

This section expands the corpus, measurement, and surface-transfer stages summarized above. All numerical panels are generated from frozen observation files.

C.1 Estimands

Table 1 separates the evaluation units used at each stage; the following material supplies the complete construction and evaluation details for those estimands.

C.2 Corpus and Measures

Proposition-conditioned corpus. Human continuations are balanced across PG-19 narrative, WikiLingua instructional text, Wikipedia encyclopedic text, and XL-Sum news, with 125 pairs from each source. Generation is balanced across MiniMax-M2.5, DeepSeek-V4-Flash, GLM-5.2, and Qwen3.6-Flash. A planner sees the natural prefix and human continuation and records 3–10 atomic propositions, required entities, and an intended endpoint. A distinct generator sees the prefix and plan but never the human continuation, and planner, generator, and judge are distinct for every pair. Acceptance requires complete or mostly complete core-content coverage, no major unsupported claim, and no major contradiction. Structural gates reject malformed, list-like, meta, refused, excessively copied, or extreme-length candidates; rejected attempts are regenerated, not edited. The resulting pairs are proposition-conditioned automatically, while semantic and rhetorical equivalence remains an empirical diagnostic.

Prefix-only corpus. For each accepted sample, the control retains its identifier, source, natural prefix, human continuation, assigned generator, and target word count. The generator receives only the prefix and scalar length request—never the plan, entity list, endpoint, human continuation, or a transformation of them. It passes the same structural gates but no content-overlap gate. This arm removes the asymmetric planning interface and has weaker semantic matching. Axes and readouts evaluated on this arm are fitted exclusively on the proposition-conditioned corpus.

Corpus ledger. The proposition-conditioned corpus contains 500 accepted pairs, balanced at 125 per domain and generator. All 500 pass the independent automatic judge and maintain distinct planner, generator, and judge models. The construction log records 50 rejected generations, including 43 rejected candidate records; accepted texts receive no post-generation repairs. The non-gating plan-serialization diagnostic flags 179 pairs, but the extreme serialization gate flags none. Mean and median machine–human length ratios are 0.935 and 0.925, with range 0.603–1.405. The proposition-conditioned semantic cosine is 0.925, the prefix-only cosine is 0.727, and within-source derangement gives 0.469. These records support controlled comparison but do not certify perfect propositional equivalence.

Measurement. The frozen inventory contains 10 lexical, 12 syntactic, and 13 discourse measurements. Seven nuisance variables—word, sentence, paragraph, and character counts, mean sentence length, named-entity rate, and number rate—are removed by training-fold regression. Feature standardization, nuisance fitting, surface-axis construction, and hidden readout fitting all occur inside the relevant training fold. The discourse group remains an off-target audit rather than a stable hidden target. Section C.3 gives the dictionary, strongest effects, complete environment holdouts, plan-removal control, and PCA sensitivity.

Feature inventory. The 10 lexical coordinates are function-word, pronoun, contraction, repeated-trigram, abstract, evaluative, emphasis, and institutional-lexicon rates, MATTR-50, and mean content-word Zipf frequency. The 12 syntactic coordinates are mean dependency depth and length;

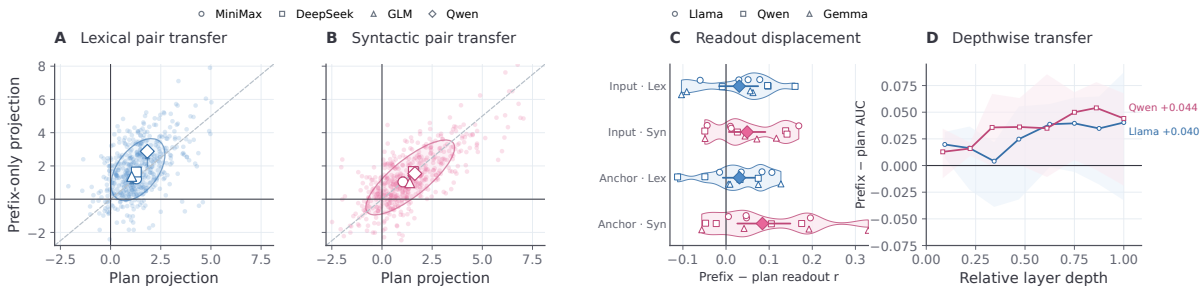


Figure 7: **Surface directions transfer after removing the proposition plan.** (A–B) Pair projections. (C) Readout-correlation changes at input and anchor layers. (D) Source-separation changes across depth. Prefix-only data are held out from fitting.

clause, coordination, and subordination densities; passive-clause, nominalization, fragment, apposition, and parenthetical rates; sentence-length coefficient of variation; and adjacent POS parallelism. The 13 discourse coordinates cover six connective families, explicit relations, transition openings, enumeration and summary markers, entity and subject continuity, and adjacent content overlap. Lexical and syntactic groups are hidden-state targets; discourse is an off-target audit. Every training fold residualizes word, sentence, paragraph, and character counts, mean sentence length, named-entity rate, and number rate.

C.3 Surface Realizations

Linguistic interpretation. The multivariate projection corresponds to recurrent, readable choices. Across 500 matched pairs, *however* appears in 83 machine documents and 34 human documents; *these* in 116 and 69; *during* in 78 and 49; *despite* in 36 and 7; and *following* in 35 and 14. Evaluative forms also separate the sources: *crucial* and *significant* each occur in 22 machine documents but only 3 human documents. The reporting frames *stated that* and *noted that* likewise occur in 16 and 13 machine documents and in no matched human document. These counts describe this corpus; none is an authorship criterion.

Distributed effects. The largest standardized shifts span both confirmatory groups: function-word rate decreases by $d = -1.221$, pronoun rate by -0.770, and mean content-word frequency by -0.759; lexical diversity increases by $+0.659$, while mean dependency length and sentence-length variation decrease by -0.654 and -0.491. Their distribution across measurements motivates a multivariate signature rather than a single-cue detector.

Paired evidence. The same tendency appears as rhetorical repackaging within proposition-conditioned pairs. A human news continuation begins, "In a statement, RBS chief executive Ross McEwen said ...," whereas its machine mate writes, "RBS chief executive Ross McEwen stated that ...." A human narrative coordinates survival with "Yet he stood on his feet," while the machine makes the relation explicit: "Despite this grievous injury, he pressed on." An instructional sentence changes from "You should consider if the beginning section flows well" to "It is crucial to check whether the beginning flows naturally." The examples instantiate the measured shifts toward explicit transitions, nominal or evaluative framing, and more regular clause packaging. They are concordances for interpreting the frozen measurements.

Figure 7 separates pair-level transfer from hidden readout transfer. Most continuations remain in the machineward quadrant at both interfaces, while the paired scatter exposes which samples move toward or across the decision origin. Across the input-to-anchor displacements, proposition-conditioned and prefix-only readout correlations are related but not interchangeable. The depth profiles show that removing the plan usually improves source separation while retaining visible generator variation.

Lexical and syntactic directions remain stable under generator holdout. Their domain-minimum cosines are lower, at 0.934 and 0.546, but remain positive. On prefix-only outputs, the original-axis lexical and syntactic cosines are 0.885 and 0.869, with positive projections for 88.0% and 79.6% of pairs. The discourse direction collapses in its weakest environments and under aggressive PCA truncation and remains an off-target measurement. On the prefix-only corpus, lexical and syntactic readouts remain predictive even though the generator receives no human-derived proposition plan.

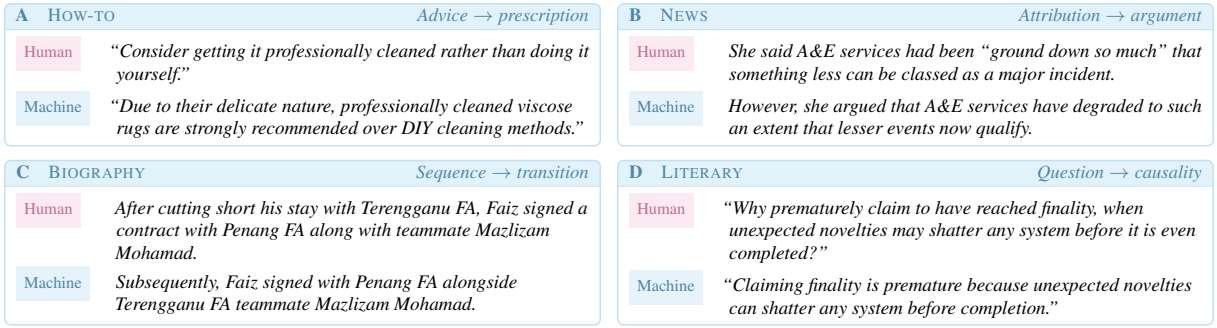


Figure 8: **Rhetorical repackaging across domains.** Complete paired excerpts illustrate prescription, argument, explicit transition, and causality; none enters model fitting.

C.4 Lexical Concordance

Lexical inventory. The more conspicuous connective choices sharpen the corpus-level description: *consequently* occurs in 21 machine documents and 1 human document, *furthermore* in 11 and 1, *additionally* in 15 and 5, and *ultimately* in 11 and 1. Evaluative closure shows the same asymmetry: *essential* occurs in 16 machine documents and 2 human documents, and *profound* in 7 and 0. The full frame *it is crucial* occurs in 10 machine continuations and no human continuation.

Corpus specificity. Popular shibboleths do not necessarily separate a matched corpus: *delve* occurs in neither source, *tapestry* occurs once in each, and *landscape* occurs in 6 machine versus 5 human documents. The recognizable contrast lies in the combination of explicit linking, evaluative closure, and regularized clause packaging.

Paired excerpts. The four readings below show how corpus-level effects appear in individual pairs. Instructions move from personal advice to impersonal prescription; news replaces repeated *said* with differentiated reporting predicates; encyclopedic prose adds explicit stage and evaluation markers; and literary rhetorical questions become declarative causal relations. Across domains, the recurring pattern is rhetorical regularization.

Same-token field. We also project 2.44–2.48M continuation-token states per model onto leave-one-generator-out nuisance-nullspace directions. An exact token is compared only in matched pairs containing it on both sides, after subtracting each continuation's mean coordinate; source separation and unequal frequency are removed from the local contrast. Of 612–704 eligible contrasts per model, 89–379 survive BH-FDR and generator/domain sign-agreement gates. The field is specific to each model and tokenizer and is estimated from hidden-state projections rather than direct-logit-lens decoding.

D Readout and Pullback

Readout. For source side a, let H_a be centered, nuisance-residualized hidden states in the frozen within-class nullspace and z_{ga} the corresponding centered surface scores. Equation 1 defines the forward covariance activation pattern and the source-consensus readout. Human and machine patterns must agree positively before their normalized sum. The reported fold statistic averages the two held-out Pearson correlations; Table 2 retains both source-side values.

Pullback. Equation 2 maps each late readout through the layer-to-layer Jacobian, averages source-side pullbacks, and projects them through the frozen nuisance projector P_m. Neither source-side vector is sign-flipped, so the declared $+w$ pole is inherited from the machine-oriented readout. Equation 3 defines the fixed-sequence response: signed τ gives directionality, while $|\tau|$ gives capacity comparable with unoriented null targets.

Matched null. Each replicate permutes surface scores within corpus–generator strata, refits every layerwise readout, refreezes the global direction, repeats the Jacobian pullback, and evaluates the identical hashed trajectories. The learned and refitted-null capacities are defined in Equation 3. We use $n = 100$, $R = 100$, five intermediate layers, and a depth-normalized curve statistic. Injection and fitted readout-anchor layers remain descriptive construction-internal diagnostics.

Figure 9 compares construction alternatives, nuisance ranks, and fit counts for all four model–axis cases.

The canonical pullback exceeds upstream encoding, raw source contrast, nullspace source con-

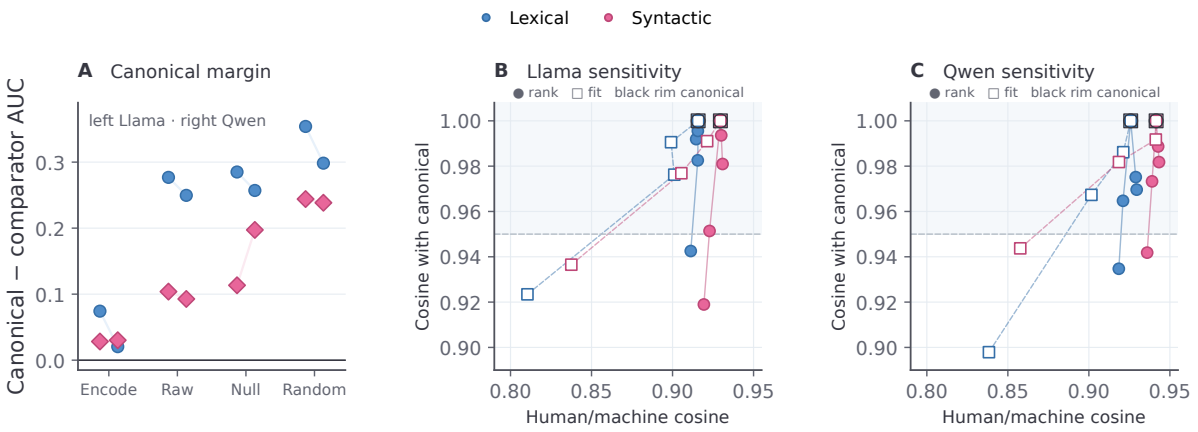


Figure 9: **Canonical pullback is stable across construction choices.** (A) Capacity margin over four alternatives. (B–C) Nuisance-rank and fit-size variants; black rings mark the frozen setting.

trast, and the projected-random ensemble in all four model–axis cases. The margin over upstream encoding is smaller than the margins over the other controls, consistent with related but distinct encoding and pullback geometry. All nuisance-rank and fit-count variants remain close to the frozen direction. Figure 3 contains the fixed-token trajectories and matched-null distributions.

E Generation Evidence

This section separates the held-out lexical test from the earlier low-dose audit and records the inference and artifact boundary.

E.1 Held-Out Lexical Test

Protocol. Hooks begin after prompt prefill. The canonical Llama lexical pullback is injected at L7 with mean pooling, and $+w$ retains the machine-oriented pole fixed by the readout. Prompt indices 0–19 form a development set used to select $\alpha = 2$, or 10% of the mean hidden-state norm. After this choice, the intervention site, pole, strength, 128-new-token budget, temperature 0.7, top-$p = 0.95$, and base seed 8236 are frozen. The formal test uses the untouched prompt indices 34–199, giving 166 prompts and three paired decode seeds.

Inference. The prompt is the unit of inference. We compute the paired $+2w/-2w$ difference within each seed, average the three differences within prompt, and bootstrap the 166 prompt means. Target projection, aggregate component selectivity, and the two target-minus-off-target contrasts use this same hierarchy; sign-flip tests also operate on the paired prompt means. Repeated decoding refines a prompt-level contrast without treating seeds or tokens as independent observations.

Result. Lexical target projection is $0.269\,[0.127, 0.412]$ with paired $d = 0.280$ and $p = 0.0006$; component selectivity is $0.287\,[0.117, 0.459]$ with $d = 0.255$ and $p = 0.0012$. The target-minus-syntax contrast is $0.234\,[0.011, 0.457]$, and target-minus-discourse is $0.340\,[0.184, 0.496]$. Target and selectivity seed means are respectively 0.123/0.267/0.416 and 0.090/0.300/0.471, so both estimands retain the declared sign in every seed.

Prompt variation. Figure 10 retains all 166 prompt coordinates after averaging within prompt across seeds. Target projection and component selectivity are strongly related at the prompt level, while their marginal distributions contain substantial variation on both sides of zero. The positive cluster means and pairwise intervals show that this heterogeneity does not depend on a few seed-level summaries.

E.2 Dose Sensitivity

Before the held-out test, the 5% protocol was evaluated across both models and both confirmatory axes. Table 4 retains those results as a sensitivity analysis. At this lower dose, target movement appears most clearly for Llama syntax, while every component-selectivity interval spans zero. The pattern motivates treating intervention magnitude as a calibrated part of the controller rather than transferring one scale across model–axis conditions.

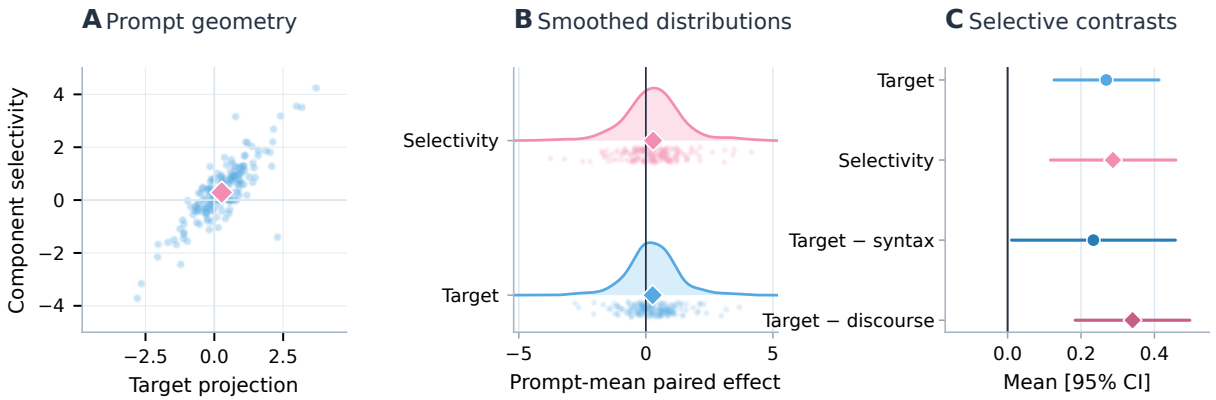


Figure 10: **Held-out lexical effects across prompts.** (A) Joint prompt geometry. (B) Smoothed marginals. (C) Target and selectivity contrasts with 95% intervals.

Model	Axis	Target [95% CI]	Selectivity [95% CI]
Llama 3.1 8B	Lex.	0.123 [−0.010, 0.255]	0.139 [−0.049, 0.345]
	Syn.	0.249 [0.043, 0.518]	0.152 [−0.092, 0.445]
Qwen3 4B	Lex.	0.012 [−0.169, 0.195]	0.027 [−0.181, 0.234]
	Syn.	0.025 [−0.189, 0.295]	0.029 [−0.221, 0.315]

Table 4: **Low-dose sensitivity.** Prompt-clustered three-seed effects at the 5% endpoint.

Guardrails. The 20-prompt 10% development run has paired-output semantic similarity $0.855\,[0.809, 0.898]$, with minimum 0.637; its target and selectivity intervals are wide because it is used for selection rather than final inference. Under the broader 5% audit, paired semantic similarity ranges from 0.864 to 0.875 across the four model–axis conditions, and most content, length, repetition, and fragmentation intervals span zero. These diagnostics support the development choice, but formal 10% semantic equivalence is not a claim of the held-out experiment.

E.3 Reproducibility

Artifact record. The artifact records dataset revisions and hashes, accepted and rejected attempts, prompt indices, model/layer conventions, direction hashes, random seeds, strengths, and per-prompt observations. English measurements use spaCy 3.8.7 with `en_core_web_sm` 3.8.0. Every fold fits nuisance regression, standardization, surface directions, and hidden readouts on training data only. Matched nulls preserve source, domain, generator, hidden states, fitting steps, and fixed-token trajectories while breaking target correspondence. Primary statistics and figures regenerate from frozen observations, and the server snapshot records local hardware and software versions.

Recorded stages total approximately 3.392 GPU-process hours. Pullback and null stages use 32 paired Jacobians and 100 pipelines × 100 sequences per concept; the held-out lexical test adds 498 paired prompt–seed observations and 127,476 generated tokens. The two models run concurrently in earlier stages, so wall time is lower; hosted corpus-generation APIs do not expose provider-side accelerator time. Table 5 gives rounded model-level time and tokens.

Model	Stage	GPU-process h	Tokens
Llama 3.1 8B-Instruct	Pullback	0.006	–
	Matched null	0.416	–
	Response + controls	0.857	424,960
	Three-seed replication	0.296	254,976
	Held-out lexical test	0.047	127,476
Qwen3 4B	Pullback	0.004	–
	Matched null	0.313	–
	Response + controls	1.116	424,960
	Three-seed replication	0.338	254,976

Table 5: **Recorded confirmatory compute.** GPU-process hours use one model-owning CUDA process; displayed rows are independently rounded.